

Retail Risk Intelligence Center

Microsoft Extended Contoso Retail Datasets

An End-to-End Data Analytics Solution

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Tools & Technologies
Python | SQL Server | Azure SQL Database | Power BI | DAX

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1. Executive Summary

The Retail Risk Intelligence Center was developed to provide a proactive approach to business performance monitoring using the Microsoft Extended Contoso retail dataset. Traditional business intelligence dashboards often focus on historical performance metrics, explaining what has happened in the past. This project was designed to move beyond descriptive reporting by identifying and monitoring operational and financial risks that could negatively impact future business performance.

The solution integrates data from multiple business functions, including sales, inventory management, customer satisfaction, and order fulfilment. Data was extracted from the [Extended Contoso Dataset](#), transformed using Python, stored in SQL Server, and modelled in Power BI using a constellation schema architecture design.

Four primary risk domains were developed:

- Revenue Risk
- Inventory Risk
- Customer Risk
- Fulfilment Risk

These risk measures were combined into an Overall Risk Score to provide executives with a consolidated view of business performance and potential operational threats.

The final solution consists of four interactive dashboard pages:

- Executive Risk Command Center
- Revenue Risk Center
- Inventory Risk Center
- Customer & Service Risk Center

The dashboard enables decision-makers to identify high-risk stores, products, inventory conditions, customer satisfaction issues, and fulfilment bottlenecks through dynamic visualisations, root-cause analysis, and executive summaries.

The project demonstrates practical application of data analytics, data modelling, SQL validation, DAX measure development, business intelligence reporting, and data storytelling techniques commonly required in modern Data Analyst roles.

2. Business Problem

Retail organisations often monitor historical performance metrics but lack visibility into operational and financial risks that may impact future performance. Management requires a centralised solution capable of identifying, measuring, and monitoring key business risks

across revenue generation, inventory management, customer satisfaction, and fulfilment operations.

3. Business Question

What risks could prevent Contoso from achieving its operational and financial objectives?

4. Project Objectives

- Identify revenue-related risks.
- Monitor inventory exposure and stock health.
- Measure customer satisfaction risk.
- Evaluate fulfilment performance.
- Provide actionable insights through interactive dashboards.
- Support management decision-making through risk-based analytics.

5. Data Collection & Architecture

5.1 Data Source

The project utilised the [Microsoft Contoso Retail Dataset](#), which contains transactional, inventory, customer, marketing, and operational data across multiple retail business functions. The dataset was provided in Parquet format and consisted of both fact and dimension tables.

The primary datasets included:

- | | |
|----------------------|-----------------------|
| • StoreSales | • Channel |
| • SalesQuota | • Currency |
| • Returns | • Customer |
| • OrderPayment | • CustomerAcquisition |
| • OrderFulfillment | • Date |
| • OnlineSales | • Employee |
| • MarketingSpend | • PaymentMethod |
| • Inventory | • Product |
| • CustomerSurvey | • Promotion |
| • ExchangeRate | • ReturnReason |
| • AcquisitionChannel | • Store |

5.2 Data Preparation Process

The source files were initially provided in Parquet format. Since SQL Server does not natively support direct Parquet imports, Python was used to convert the files into CSV format.

5.3 Data Storage

After converting the source Parquet files into CSV format, all datasets were imported into Azure SQL Database. Azure SQL served as the central data repository for this project, providing a structured environment for data exploration, validation, and preparation before connecting the data to Power BI.

Rather than importing the data directly into Power BI, I first used SQL Server to understand the dataset, verify data quality, and identify any issues that could affect the reliability of the analysis.

The validation process included:

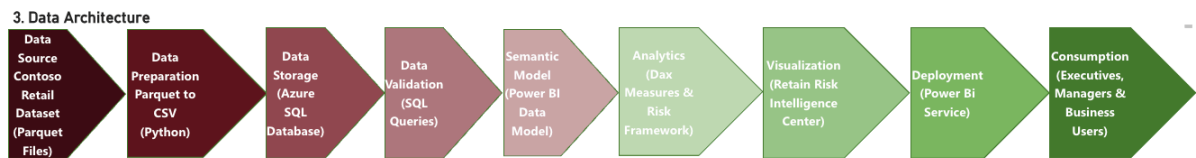
- Verifying that all datasets were successfully imported into the database.
- Exploring the structure and business purpose of each table.
- Validating relationships between fact and dimension tables.
- Checking for missing or unmatched records that could affect reporting accuracy.
- Confirming that a single CurrencyKey existed across the transactional datasets, eliminating the need for currency conversion.
- Investigating ProductKey records that did not appear within transactional tables.
- Examining inventory records to determine whether the data represented inventory snapshots or inventory transactions.
- Assessing the availability and geographic coverage of customer survey responses.
- Reviewing date ranges across business processes to support time-based analysis.
- Evaluating the suitability of each dataset for KPI development and risk calculations.
- Identifying potential limitations and assumptions that needed to be considered during dashboard development.

These activities helped me gain a deeper understanding of the underlying business processes and informed the design of the final Power BI data model, KPI framework, and Retail Risk Intelligence Center dashboard.

5.4 Analytical Architecture

The analytical architecture was designed to support an end-to-end data analytics workflow, transforming raw retail data into actionable business intelligence. The solution integrates multiple technologies, including Python, Azure SQL Database, SQL, Power BI, and DAX, to support data preparation, validation, modelling, analysis, visualisation, and deployment.

The architecture follows a layered approach in which data flows from source systems through a series of processing and analytical stages before being delivered to business users through interactive dashboards.



At the data preparation stage, Python was used to convert the source Parquet files into CSV format. The processed datasets were then stored in Azure SQL Database, where SQL queries were used to explore the data, validate quality, investigate business processes, and assess the suitability of the data for analytical reporting.

Following validation, the datasets were imported into Power BI and transformed into a constellation schema consisting of multiple fact tables and shared dimensions. This semantic model provided the foundation for KPI development and risk analysis.

DAX measures were subsequently developed to calculate business metrics, risk indicators, and the Overall Risk Score framework. These calculations transformed operational data into meaningful analytical insights.

The final dashboards were developed within Power BI Desktop and published to Power BI Service, enabling centralised access and interactive consumption by business users and decision-makers.

This architecture ensured a structured, scalable, and repeatable analytical workflow capable of supporting business performance monitoring and proactive risk management.

The architecture provides an end-to-end analytics solution, demonstrating the complete workflow from raw data acquisition through to business intelligence delivery and consumption.

6. Data Understanding

6.1. Dataset Overview

The Contoso dataset used in this project consisted of 22 datasets representing various retail business processes, including sales, inventory management, customer acquisition, marketing, customer satisfaction, order fulfilment, and supporting reference data. The datasets were categorised into fact tables containing transactional or performance-related data and dimension tables containing descriptive business attributes.

Fact Tables

Table Name	Type	Description
StoreSales	Fact	Physical store sales transactions
OnlineSales	Fact	Online sales transactions
Inventory	Fact	Inventory snapshot records
SalesQuota	Fact	Revenue targets by store and period
OrderFulfillment	Fact	Delivery performance and SLA records
CustomerSurvey	Fact	Customer satisfaction survey responses
CustomerAcquisition	Fact	Customer acquisition performance data

Table Name	Type	Description
MarketingSpend	Fact	Marketing expenditure records
Returns	Fact	Product return transactions
OrderPayment	Fact	Customer payment transactions
ExchangeRate	Fact	Currency exchange rate records

Dimension Tables

Table Name	Type	Description
Product	Dimension	Product hierarchy and attributes
Store	Dimension	Store information and location details
Customer	Dimension	Customer attributes and demographics
Date	Dimension	Calendar and time intelligence attributes
Channel	Dimension	Sales channel information
Promotion	Dimension	Promotional campaign information
Currency	Dimension	Currency reference information
Employee	Dimension	Employee and workforce information
PaymentMethod	Dimension	Payment method classifications
ReturnReason	Dimension	Product return reason classifications
AcquisitionChannel	Dimension	Customer acquisition channel information

The combination of these datasets provided a comprehensive view of retail operations and enabled the development of integrated analytics across revenue performance, inventory management, customer satisfaction, fulfilment operations, customer acquisition, and marketing effectiveness. These datasets formed the foundation for the development of the Retail Risk Intelligence Center and supported the creation of risk-based KPIs, business insights, and interactive dashboard reporting.

6.2. Tables Selected for Analysis

Although the Contoso dataset contained 22 tables, not all datasets were required to address the project's business objective of developing a Retail Risk Intelligence Center. Following data exploration and validation, a subset of tables was selected for the final analytical model based on their relevance to revenue performance, inventory management, customer satisfaction, and fulfilment operations.

The primary fact tables used in the final solution were:

- StoreSales
- OnlineSales
- Inventory

- SalesQuota
- CustomerSurvey
- OrderFulfillment

The primary dimension tables used in the final model were:

- Date
- Product
- Store
- Customer

Additional tables were explored during the data understanding and validation phases, but were excluded from the final analytical model because they were not directly required to support the project's risk monitoring objectives. This approach helped maintain a simplified, efficient, and business-focused data model while ensuring that all required analytical capabilities were preserved.

6.3 Data Validation and Quality Assessment

Before data modelling and dashboard development, a comprehensive data validation process was performed to assess data quality, verify business logic, identify potential anomalies, and ensure the reliability of analytical outputs. Validation was conducted using SQL queries executed against the SQL Server database after data ingestion.

6.3.1 Currency Validation

Financial measures such as Sales Amount, Sales Quota, and Inventory Value require consistent currency reporting to ensure meaningful aggregation and comparison. A validation query was run to determine the number of distinct currency identifiers in the transactional datasets.

Validation Result

Analysis confirmed that only a single CurrencyKey existed across the primary fact tables.

Business Impact

Since all financial transactions were recorded using a single reporting currency, currency conversion and foreign exchange adjustments were not required. Consequently, all monetary values were aggregated directly throughout the analytical model.

Conclusion

Financial metrics were treated as a single reporting currency, and no exchange rate transformation was necessary.

6.3.2 Product Dimension Coverage Validation

A referential integrity assessment was conducted between the Product dimension table and transactional fact tables to verify that all products contained transactional records.

Validation Result

The Product dimension contained ProductKey 944, which did not appear within any transactional fact table.

Business Impact

The missing transaction activity indicated that the product existed as a master record but was not actively sold or recorded within the analysed period.

Conclusion

The product record was retained within the model because it did not affect analytical calculations or relationship integrity.

6.3.3 Inventory Structure Validation

The Inventory dataset required additional investigation to determine whether records represented inventory transactions or inventory snapshots. Several inventory records were analysed across products, stores, and reporting dates.

Validation Result

Inventory values represented periodic inventory snapshots rather than inventory movement transactions. The dataset captured the inventory position at specific points in time, including inventory quantity, inventory value, stock status, and reorder indicators.

Business Impact

Traditional cumulative inventory calculations would produce inaccurate results because inventory records were not transactional.

Conclusion

Inventory measures were designed using snapshot-based logic and latest-date calculations to accurately reflect inventory positions.

6.3.4 Customer Survey Coverage Assessment

Customer satisfaction analysis was based on responses collected within the CustomerSurvey dataset. A geographic coverage assessment was conducted to determine the availability of survey responses across operating countries.

Validation Result

Customer survey records were available for only six countries despite the organisation operating across a significantly larger geographic footprint.

Business Impact

Customer Risk metrics, Net Promoter Score (NPS), and Customer Satisfaction (CSAT) measures only represent surveyed countries and should not be interpreted as a complete representation of global customer sentiment.

Conclusion

Customer Risk calculations were restricted to available survey data and this limitation was documented throughout the dashboard and project report.

6.3.5 Date Dimension Validation

Time-based analysis required a dedicated calendar table to support year-to-date calculations, trend analysis, period comparisons, and consistent filtering across all reports.

Validation Result

Instead of using the original Date dimension provided in the dataset, a new Date table was created in Power BI. The custom Date table was designed to cover the required reporting period and included additional attributes such as Year, Quarter, Month, Month Name, and Day.

Business Impact

Creating a dedicated Date table improved flexibility for time intelligence calculations and ensured consistent date filtering across all fact tables used in the dashboard.

Conclusion

A custom Date table created in Power BI was used as the central calendar dimension within the data model.

6.3.6 Relationship Validation

Fact-to-dimension relationships were validated to ensure accurate filtering behaviour and aggregation across the Power BI model.

Validation Result

Primary business dimensions, including Product, Store, Customer, and Date, successfully established relationships with the relevant fact tables. Relationship validation confirmed that these dimensions provided the necessary business context for sales, inventory, customer satisfaction, and fulfilment analysis.

During the modelling process, it was identified that CustomerSurvey and OrderFulfillment represented distinct business processes with different levels of granularity from sales transactions. As a result, these tables required separate analytical treatment while still leveraging shared dimensions where appropriate.

The relationship structure was reviewed to ensure that filtering behaviour remained consistent across the model and that analytical calculations accurately reflected the underlying business processes.

Business Impact

Appropriate relationship design prevented double-counting, ambiguous filtering paths, and inaccurate aggregations. The use of shared dimensions enabled consistent analysis across multiple business functions while maintaining separation between independent operational processes.

Conclusion

The final data model successfully supported a constellation schema architecture consisting of multiple fact tables connected to shared business dimensions. This structure provided a scalable and efficient foundation for KPI development, risk calculations, and enterprise-level reporting within the Retail Risk Intelligence Center.

6.3.7 Data Quality Summary

The validation process confirmed that the dataset was suitable for analytical reporting and dashboard development.

Key findings included:

- Single reporting currency across all financial datasets.
- Inventory data stored as periodic snapshots.
- One inactive product record identified within the Product dimension.
- Customer survey coverage limited to six countries.
- Custom Date table created and used for time intelligence calculations.
- Valid relationships established across the analytical model.
- No critical data quality issues identified that would materially impact dashboard results.

The outcomes of these validation activities informed the design of the data model, KPI framework, and risk calculations implemented within the Retail Risk Intelligence Center.

7. Data Modelling

7.1 Data Modelling Approach

Following data preparation and validation, the datasets were transformed into an analytical data model within Power BI to support integrated reporting and risk analysis. The primary objective of the model was to provide a unified analytical framework capable of supporting multiple business processes, including revenue performance, inventory management, customer satisfaction, and order fulfilment.

A dimensional modelling approach was adopted due to its simplicity, scalability, and suitability for business intelligence reporting. The model was designed to separate transactional data from descriptive business attributes, enabling efficient filtering, aggregation, and analytical calculations.

The final model follows a **constellation schema architecture (fact constellation)**, where multiple fact tables share common dimension tables. This approach enables different business processes to be analyzed independently while maintaining consistency across reporting dimensions such as products, stores, customers, and time periods.

The model serves as the foundation for all KPI calculations, risk measures, and dashboard visualisations developed within the Retail Risk Intelligence Center.

7.2 Fact Tables

Fact tables contain measurable business events and quantitative data used for analytical calculations.

StoreSales

The StoreSales fact table contains physical retail store transactions and serves as the primary source for revenue-related analysis. The table includes measures such as sales amount, quantity sold, cost, profit, and discount values.

OnlineSales

The OnlineSales fact table captures e-commerce transactions and provides visibility into online sales performance. Similar to StoreSales, it contains transactional measures used for revenue and profitability analysis.

Inventory

The Inventory fact table stores inventory snapshot records representing inventory positions at specific points in time. The table contains inventory quantities, inventory values, stock status classifications, safety stock information, and reorder indicators. This table forms the basis of inventory risk calculations.

SalesQuota

The SalesQuota fact table contains target revenue values assigned to stores and reporting periods. The table supports quota attainment analysis and revenue risk calculations by enabling comparisons between actual sales and planned performance.

CustomerSurvey

The CustomerSurvey fact table contains customer satisfaction survey responses, including Net Promoter Score (NPS), Customer Satisfaction Score (CSAT), recommendation

indicators, and satisfaction status. These measures support customer experience and customer risk analysis.

OrderFulfillment

The OrderFulfillment fact table captures operational delivery performance information, including fulfilment status, processing time, delivery time, SLA compliance, shipping methods, and delay root causes. The table provides the foundation for fulfilment risk assessment.

7.3 Dimension Tables

Dimension tables provide descriptive business context and support filtering, grouping, and segmentation within analytical reports.

Date

The Date dimension was created within Power BI to support time intelligence calculations, trend analysis, period comparisons, and consistent date filtering across all fact tables. The table contains attributes such as Year, Quarter, Month, Month Name, and Date.

Store

The Store dimension contains store-level attributes including store name, city, region, country, store type, and operational characteristics. This dimension enables geographic and store-level performance analysis.

Product

The Product dimension provides product hierarchy information including product name, category, subcategory, manufacturer, and product attributes. This dimension supports product-level analysis across sales and inventory processes.

Customer

The Customer dimension contains customer attributes used for segmentation and customer-related analysis. This includes demographic and behavioural information used to evaluate customer satisfaction and acquisition performance.

7.4 Relationship Design

Relationships were established between fact tables and shared dimensions using primary and foreign key fields. One-to-many relationships were implemented wherever possible to ensure efficient filtering behaviour and accurate aggregation.

The Date dimension acts as a shared calendar dimension, enabling consistent time-based analysis across multiple business processes. Similarly, the Store and Product dimensions provide common analytical perspectives across sales, inventory, and operational datasets.

CustomerSurvey and OrderFulfillment represent distinct business processes that operate at different levels of granularity from sales transactions. As a result, these tables were modelled independently while utilising shared dimensions where appropriate. This design preserves analytical accuracy and prevents ambiguity in calculations.

The relationship structure was carefully validated to ensure:

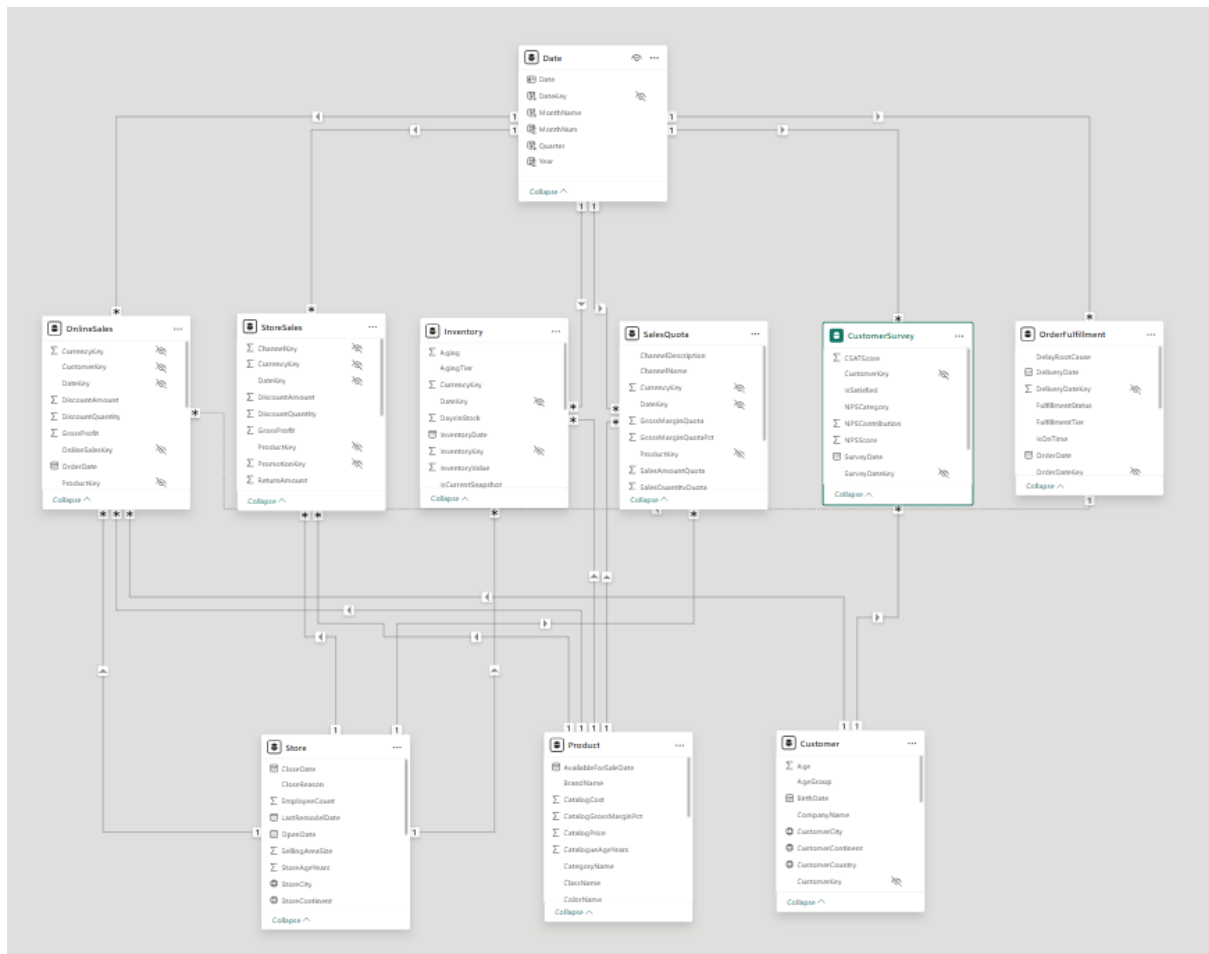
- Accurate filter propagation
- Consistent business definitions
- Elimination of ambiguous relationships
- Prevention of double counting
- Reliable KPI calculations

This design supports integrated analysis while maintaining clear separation between individual business processes.

7.5 Constellation Schema Architecture

The final analytical model was implemented using a constellation schema architecture consisting of multiple fact tables connected to shared business dimensions.

Constellation Schema Architecture



Unlike a traditional star schema, which contains a single fact table, the constellation schema enables multiple business processes to coexist within the same analytical environment. Shared dimensions such as Date, Store, Product, and Customer provide consistency across all reporting domains while allowing each business process to retain its own level of detail.

This architecture supports the integration of:

- Revenue Performance Analysis
- Inventory Risk Analysis
- Customer Satisfaction Analysis
- Fulfilment Performance Analysis

within a single reporting solution.

The final model provides a scalable, maintainable foundation for dashboard development and supports all risk calculations implemented within the Retail Risk Intelligence Centre.

7.6 Benefits of the Data Model

The implemented data model delivered several analytical and operational benefits:

- Improved report performance through dimensional modelling principles.
- Consistent filtering across multiple business processes.

- Simplified KPI development and DAX calculations.
- Reduced risk of data duplication and aggregation errors.
- Enhanced scalability for future analytical requirements.
- Support for integrated enterprise-wide risk monitoring.

The model successfully transformed raw operational data into a structured analytical framework capable of supporting strategic decision-making and proactive risk management.

8. KPI Framework and Risk Methodology

8.1 KPI Development Approach

The primary objective of this project was to identify potential risks that could negatively impact organizational objectives. To achieve this, a KPI framework was developed that transformed operational and transactional data into measurable risk indicators.

The KPI framework was designed around four core business risk domains:

- Revenue Risk
- Inventory Risk
- Customer Risk
- Fulfillment Risk

Each risk domain was calculated independently and then combined into an Overall Risk Score to provide executives with a consolidated view of business risk exposure.

8.2 Revenue Performance KPIs

Total Sales

Total Sales represents the combined sales revenue generated from physical store transactions and online sales channels.

Business Purpose

Total Sales provides the baseline measure of business performance and serves as the foundation for revenue risk calculations.

Sales Quota

Sales Quota represents the revenue targets assigned to stores and reporting periods.

Business Purpose

Sales Quotas provide benchmark performance expectations against which actual sales performance can be evaluated.

Quota Attainment %

Quota Attainment measures the proportion of revenue targets achieved.

Formula:

$$\text{Quota Attainment \%} = \text{Total Sales} \div \text{Sales Quota}$$

Business Purpose

This KPI indicates how successfully the organisation is meeting planned revenue objectives.

Revenue Risk %

Revenue Risk measures the likelihood of failing to achieve revenue targets.

Formula:

$$\text{Revenue Risk \%} = 1 - \text{Quota Attainment \%}$$

Business Purpose

Higher Revenue Risk values indicate increasing gaps between actual performance and planned revenue expectations.

8.3 Inventory Management KPIs

Inventory Value

Inventory Value represents the total financial value of inventory available across all stores and products.

Business Purpose

This KPI provides visibility into the capital invested in inventory assets.

Inventory at Risk Value

Inventory at Risk Value measures the financial value of inventory classified as operationally risky.

Inventory considered at risk includes:

- Low Stock
- Stock-Out
- Overstocked
- Severely Overstocked

Business Purpose

This KPI identifies inventory exposure that may result in lost sales, increased carrying costs, or operational inefficiencies.

Inventory Risk %

Inventory Risk measures the proportion of inventory value classified as operationally risky.

Formula:

$\text{Inventory Risk \%} = \text{Inventory at Risk Value} \div \text{Inventory Value}$

Business Purpose

Higher Inventory Risk values indicate increased inventory imbalance and greater operational exposure.

8.4 Customer Experience KPIs

Net Promoter Score (NPS)

Net Promoter Score measures customer loyalty and willingness to recommend the organisation.

Formula:

$\text{NPS} = \% \text{ Promoters} - \% \text{ Detractors}$

Business Purpose

NPS is a widely accepted indicator of customer advocacy and future business growth potential.

Customer Satisfaction Score (CSAT)

CSAT measures overall customer satisfaction using survey responses.

Business Purpose

This KPI provides direct insight into customer perceptions of products and services.

Customer Risk %

Customer Risk represents the proportion of customers expressing dissatisfaction through survey responses.

Formula:

$\text{Customer Risk \%} = (\text{Number of Unsatisfied Customers} \div \text{Total Survey Responses})$

Business Purpose

Higher Customer Risk values indicate declining customer sentiment and increased risk of customer attrition.

8.5 Fulfilment Performance KPIs

SLA Breach %

SLA Breach % measures the proportion of orders that failed to meet agreed delivery service levels.

Formula:

$\text{SLA Breach \%} = (\text{Breached Orders} \div \text{Total Orders})$

Business Purpose

This KPI identifies operational performance issues that may negatively affect customer satisfaction.

Fulfilment Failure Risk %

Fulfilment Failure Risk measures the proportion of orders classified as operationally unsuccessful.

Formula:

$$\text{Fulfilment Failure Risk \%} = (\text{Failed or Delayed Orders} \div \text{Total Orders})$$

Business Purpose

Higher Fulfilment Risk values indicate increasing operational inefficiencies and customer service challenges.

8.6 Overall Risk Score

To provide executives with a single consolidated measure of business exposure, the four risk domains were combined into an Overall Risk Score.

In this project, all four risk domains were assigned equal importance. Therefore, an equal-weight methodology was used, ensuring that each risk area contributed equally to the final score.

Formula:

$$\begin{aligned} \text{Overall Risk Score} = & \\ & (\text{Revenue Risk} \times 0.25) + \\ & (\text{Inventory Risk} \times 0.25) + \\ & (\text{Customer Risk} \times 0.25) + \\ & (\text{Fulfilment Risk} \times 0.25) \end{aligned}$$

Business Purpose

The Overall Risk Score provides a unified measure of organisational risk exposure and enables management to quickly assess overall business health across multiple operational dimensions.

8.7 Risk Classification Framework

To improve interpretability, Overall Risk Scores were categorised into risk levels.

6. Risk Classification Table

Risk Score	Level of Risk
0–25%	Low
25–50%	Moderate
50–75%	High
>75%	Critical

Business Purpose

Risk classifications provide executives with an intuitive assessment of business conditions and support prioritisation of corrective actions.

8.8 KPI Framework Summary

The KPI framework transformed raw transactional, operational, and customer data into actionable business intelligence. By integrating revenue, inventory, customer satisfaction, and fulfilment performance into a unified risk model, the framework enabled proactive identification of operational vulnerabilities and supported data-driven decision-making.

The resulting metrics formed the analytical foundation of the Retail Risk Intelligence Center and were subsequently visualised through interactive Power BI dashboards.

9. Dashboard Development and Visualisation Design

9.1 Dashboard Design Approach

The Retail Risk Intelligence Center was developed as an interactive business intelligence solution designed to support executive decision-making through risk-focused analytics.

The dashboard was structured around four key business risk domains:

- Executive Risk Overview
- Revenue Risk Center
- Inventory Risk Center
- Customer & Service Risk Center

Each page was designed to answer a specific business question while maintaining a consistent visual layout, filtering experience, and reporting structure.

Common report features included:

- Interactive slicers
- Dynamic KPI cards
- Trend analysis
- Geographic analysis
- Top-risk identification
- Root-cause analysis
- Executive summaries

The dashboard enables users to move from a high-level business risk assessment to a detailed operational investigation.

9.2 Executive Risk Overview

Business Purpose

The Executive Risk Overview serves as the primary landing page of the solution and provides senior management with a consolidated view of organisational risk exposure.

The page answers the business question:

What is the current level of business risk, and where should management focus attention?

The page includes:

- Overall Risk Score
- Revenue Risk
- Inventory Risk
- Customer Risk
- Fulfillment Risk

These indicators provide an immediate assessment of overall business health.

Visualizations Used

KPI Cards

Used to display overall risk exposure and individual risk components.

Risk Trend Analysis

A line chart is used to track changes in the Overall Risk Score over time.

Risk Distribution Chart

A column chart is used to compare the relative contribution of each risk domain.

Top 10 High-Risk Stores

Bar chart used to identify stores contributing most significantly to organisational risk.

Top 10 High-Risk Products

A bar chart is used to identify products with the highest overall risk exposure.

Geographic Risk Map

Azure Maps visualisation used to display risk distribution across geographic locations.

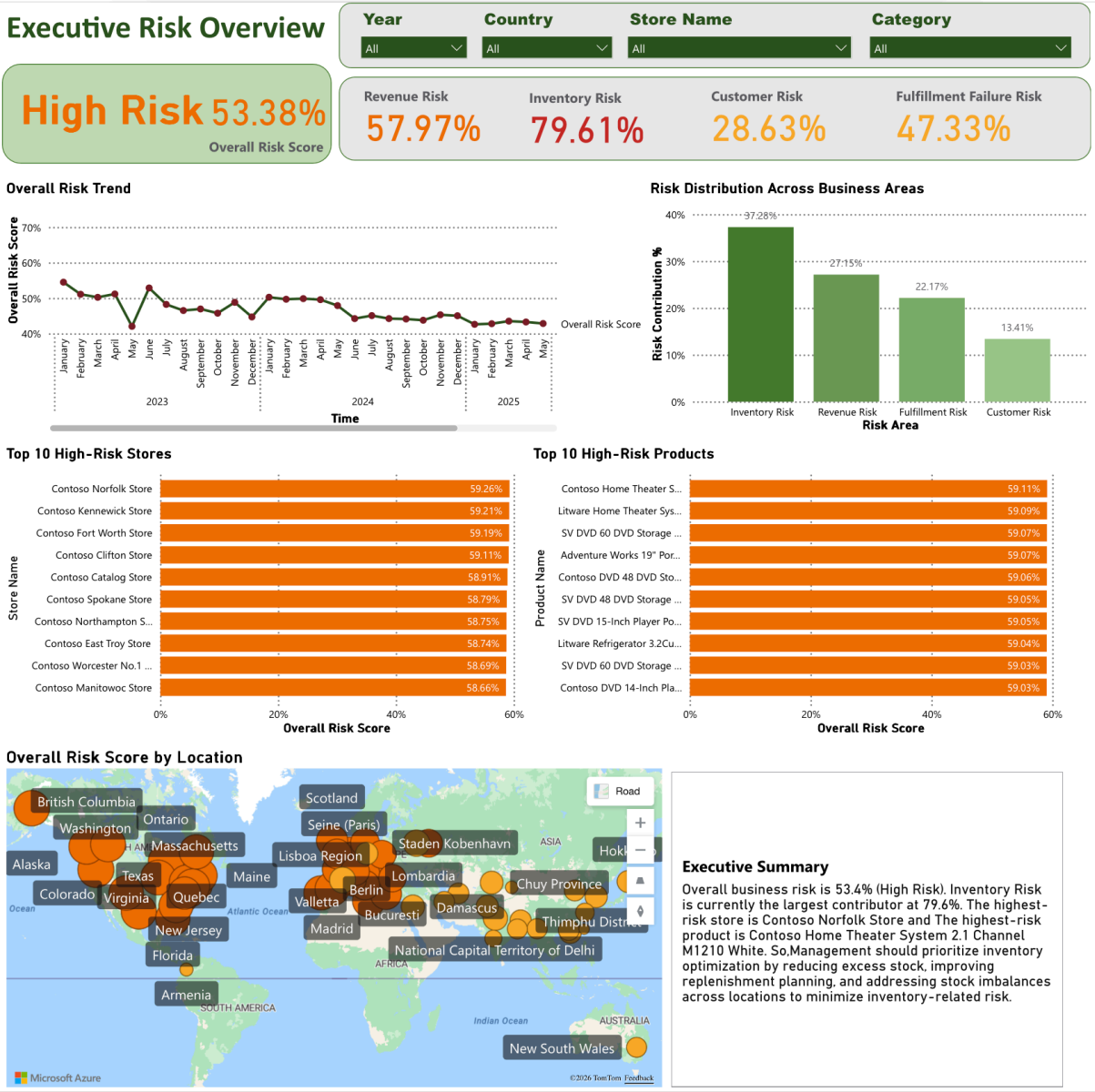
Dynamic Executive Summary

A narrative visual used to automatically summarise current business conditions and highlight key risk drivers.

Business Value

The Executive Risk Overview provides management with a centralized risk-monitoring environment that supports rapid identification of emerging issues and prioritization of corrective actions.

Executive Risk Overview Page Screenshot



9.3 Revenue Risk Center

Business Purpose

The Revenue Risk Center focuses on sales performance and revenue target attainment.

The page answers the business question:

Where is the organisation failing to achieve revenue targets, and what factors are contributing to the shortfall?

Key KPIs

- Total Sales
- Sales Quota
- Quota Attainment %
- Revenue Risk %

Visualizations Used

Revenue Performance Gauge

Displays quota attainment performance against target expectations.

Sales vs Quota Trend Analysis

Line chart comparing actual sales performance with planned targets over time.

Top 10 Underperforming Stores

Bar chart highlighting stores with the largest revenue gaps.

Top 10 Underperforming Products

Bar chart identifying products contributing most significantly to revenue shortfalls.

Revenue Root Cause Analysis

Decomposition Tree used to investigate revenue gaps across:

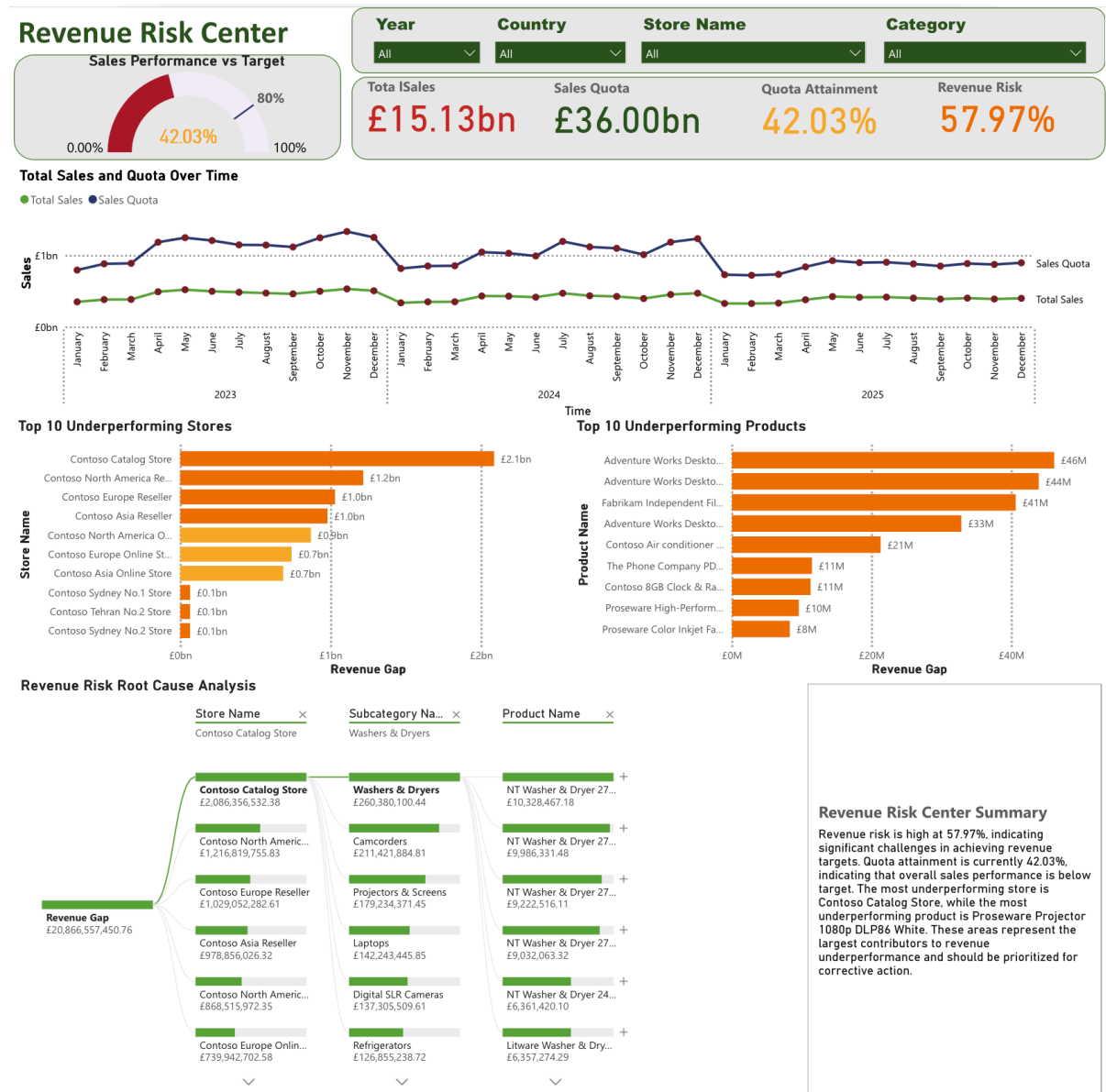
- Product Category
- Product Subcategory
- Product
- Country
- City
- Store

Dynamic Revenue Summary

Narrative visual automatically highlights major revenue performance issues and areas requiring management attention.

Business Value

The Revenue Risk Center enables management to identify underperforming business units and prioritise corrective actions to improve revenue attainment.



9.4 Inventory Risk Center

Business Purpose

The Inventory Risk Center focuses on inventory exposure and stock health management.

The page answers the business question:

Which inventory conditions present the greatest operational risk to the organisation?

Key KPIs

- Inventory Value
- Inventory at Risk Value
- Inventory Risk %

- Low Stock Count
- Stock-Out Count

Visualizations Used

Inventory Status Distribution

Donut chart displaying the distribution of inventory across stock conditions.

Inventory Risk Trend Analysis

Line chart showing changes in Inventory Risk over time.

Inventory Risk Summary Matrix

Matrix visual summarising inventory exposure across product categories, sub-categories, countries and products

Inventory Risk Treemap

Treemap visual highlighting inventory value concentration across stock conditions.

Top 10 Risky Stores

Bar chart identifying stores with the highest inventory risk.

Top 10 Risky Products

Bar chart identifying products contributing most significantly to inventory risk.

Inventory Root Cause Analysis

Decomposition Tree used to investigate inventory risk drivers across:

- Store
- Category
- Subcategory
- Product
- Location

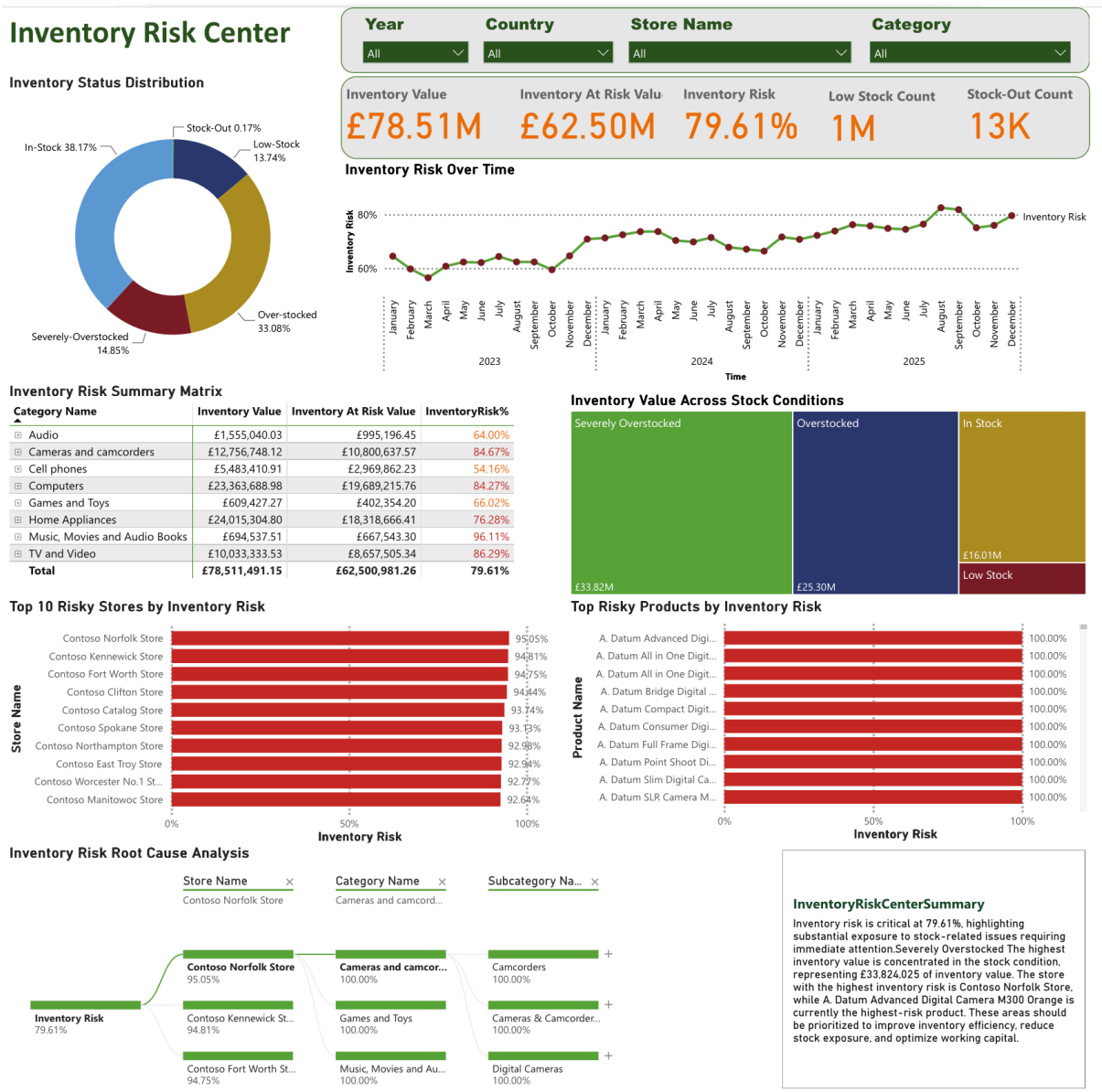
Dynamic Inventory Summary

Narrative visual summarizing inventory exposure and highlighting priority areas for intervention.

Business Value

The Inventory Risk Center supports proactive inventory management by identifying stock imbalances, inventory concentration risks, and potential supply chain inefficiencies.

Inventory Risk Center Page Screenshot



9.5 Customer & Service Risk Center

Business Purpose

The Customer & Service Risk Center evaluates customer satisfaction and operational service performance.

The page answers the business question:

How effectively is the organisation meeting customer expectations and service commitments?

Key KPIs

- Customer Risk %
- Net Promoter Score (NPS)
- Average CSAT Score

- Fulfillment Risk %
- SLA Breach %

Visualizations Used

CSAT Trend Analysis

Line chart tracking customer satisfaction over time.

NPS Trend Analysis

Line chart monitoring changes in customer advocacy.

NPS Distribution

Donut chart displaying Promoters, Passives, and Detractors.

Customer Risk by Country

Bar chart comparing customer risk across surveyed countries.

SLA Breach by Shipping Method

Column chart identifying shipping methods associated with service failures.

Orders by Delay Root Cause

Column chart highlighting operational causes of delivery delays.

Fulfilment Performance by Tier

Column chart evaluating delivery performance across fulfilment categories.

Fulfilment Root Cause Analysis

Decomposition Tree used to investigate operational service failures across:

- Delay Root Cause
- Shipping Method
- Fulfillment Tier

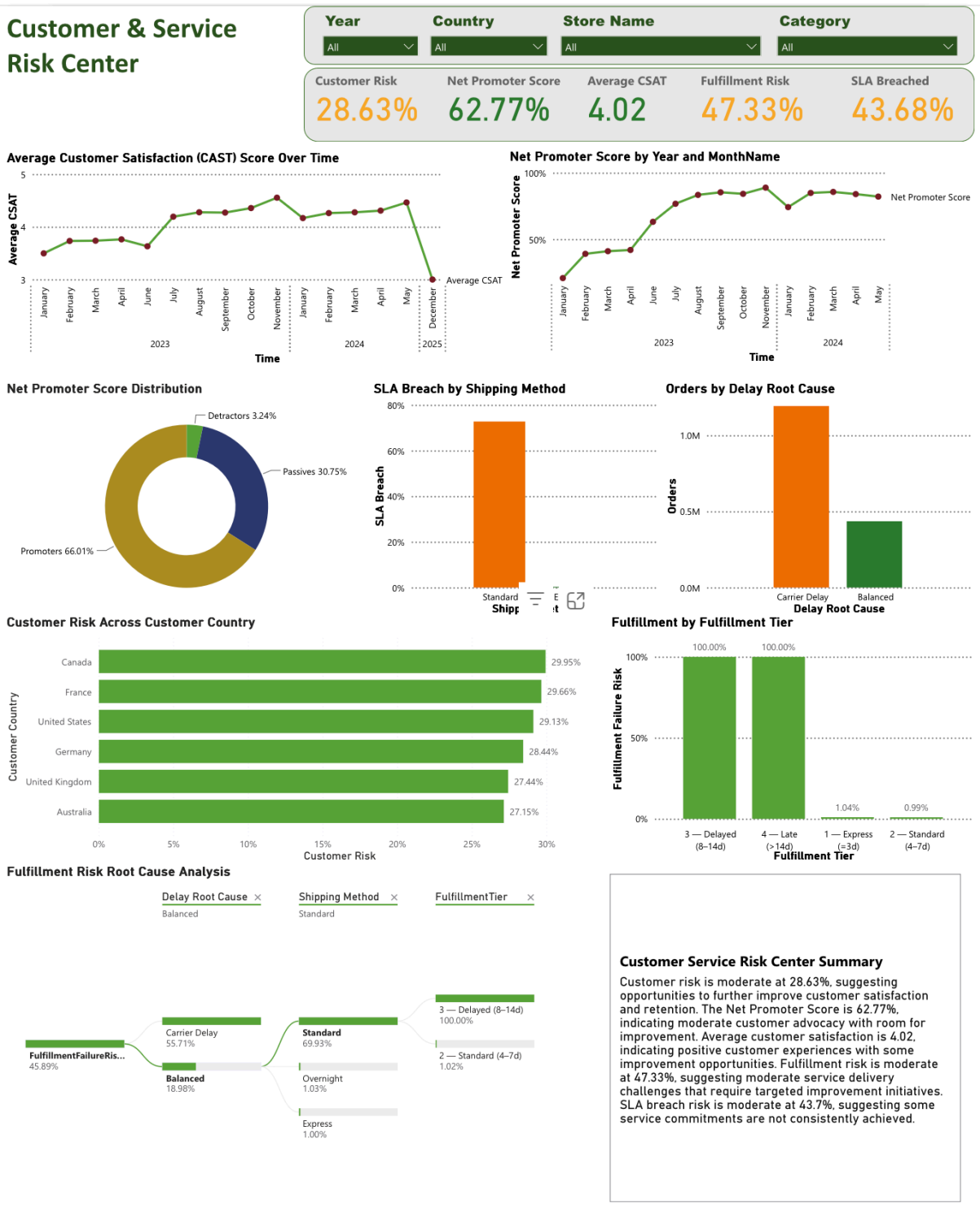
Dynamic Customer & Service Summary

Narrative visual summarising customer sentiment and operational performance.

Business Value

The Customer & Service Risk Center provides insight into customer experience and operational effectiveness, enabling management to identify service weaknesses, improve customer satisfaction, and reduce fulfilment-related risks.

Customer & Service Risk Center Page Screenshot

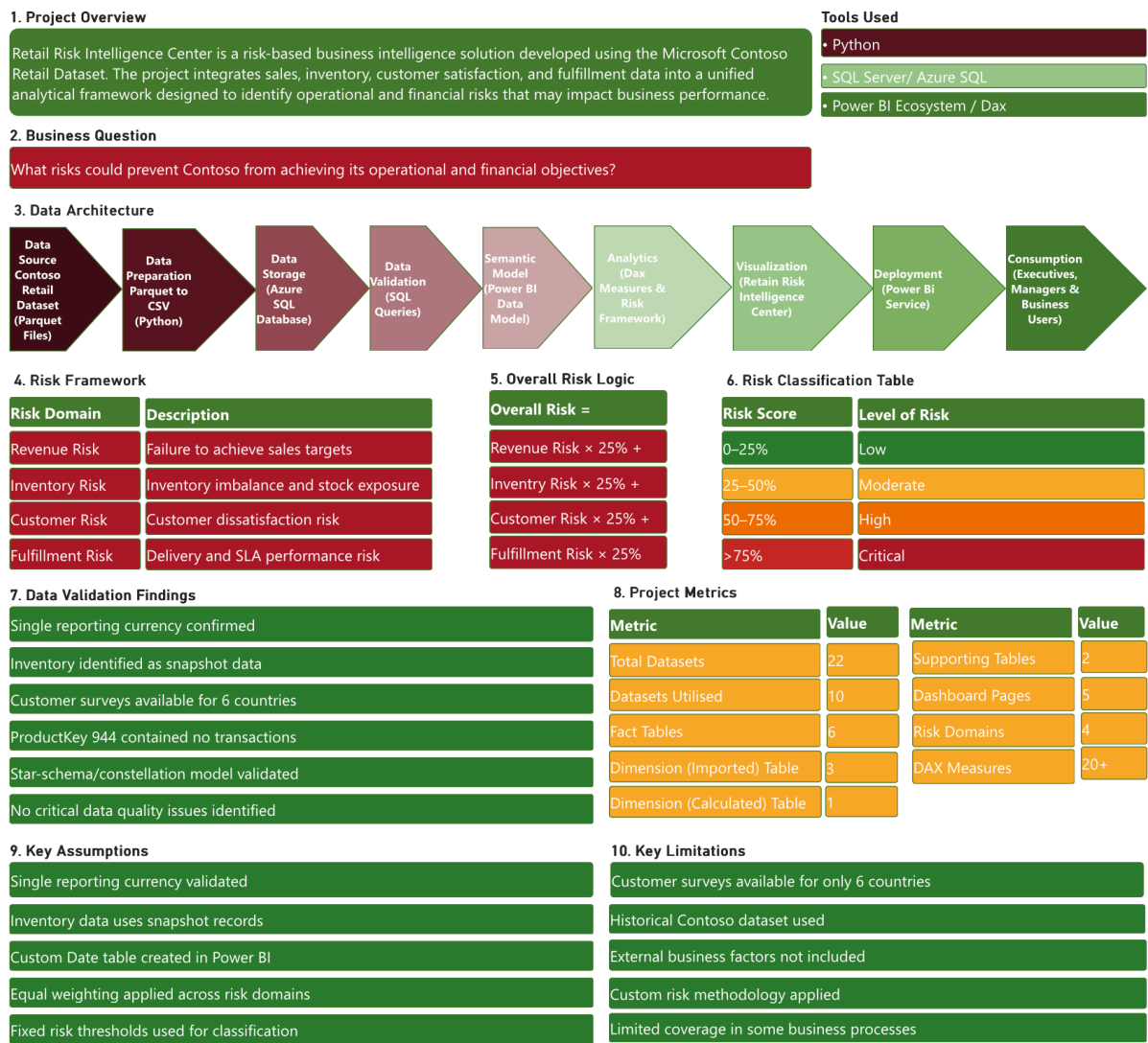


9.6 Project Documentation and Methodology

This page provides an overview of the project methodology, technical architecture, risk framework, and dashboard functionality. It summarises the end-to-end workflow from data preparation and validation to data modelling, KPI development, dashboard design, deployment, and business consumption.

Project Documentation & Methodology Page Screenshot

Project Documentation & Methodology



9.7 Dashboard Design Summary

The final dashboard solution integrates multiple business functions into a unified risk-monitoring platform. Through interactive visualisations, dynamic narratives, and root-cause analysis capabilities, the Retail Risk Intelligence Center enables users to monitor risk exposure, investigate underlying causes, and support evidence-based decision-making.

The dashboard transforms complex operational data into actionable business intelligence and provides a comprehensive view of organisational risk across revenue, inventory, customer experience, and fulfilment performance.

For an interactive experience and detailed exploration of the dashboard, access the embedded Power BI report using the link provided below.

[Embedded Dashboard Link:](#)

10. Key Findings and Business Recommendations

10.1 Overview of Findings

The Retail Risk Intelligence Center identified a number of operational and financial challenges affecting overall business performance. Analysis across revenue, inventory, customer satisfaction, and fulfilment operations revealed that the organisation is currently operating within the High Risk category, with an Overall Risk Score of 53.38%.

Inventory management emerged as the most significant contributor to organisational risk, followed by revenue performance and fulfilment operations. Customer-related risk was comparatively lower but still indicated opportunities for improvement.

The following sections summarise the key findings and associated recommendations.

10.2 Finding 1: Overall Business Risk is High

Observation

The Overall Risk Score was calculated at 53.38%, placing the organization within the High Risk category according to the established risk classification framework.

This result indicates that multiple business functions are experiencing performance challenges simultaneously, increasing overall organisational exposure.

Business Implications

A high overall risk level suggests that current operational performance may negatively impact profitability, customer satisfaction, inventory efficiency, and future business growth if corrective actions are not implemented.

Recommendation

Management should prioritize the highest contributing risk areas and establish a structured risk monitoring process. Regular review of dashboard KPIs should be incorporated into operational planning and performance management activities.

10.3 Finding 2: Inventory Risk is the Largest Risk Driver

Observation

Inventory Risk was calculated at 79.61%, making it the largest contributor to the Overall Risk Score.

Analysis of inventory status distributions revealed significant inventory concentration within overstocked and severely overstocked categories, while low-stock and stock-out conditions were also present.

Several stores and products exhibited exceptionally high inventory risk levels.

Business Implications

Excess inventory increases storage costs, ties up working capital, and raises the likelihood of product obsolescence. Conversely, stock-outs and low-stock conditions may result in lost sales opportunities and reduced customer satisfaction.

The coexistence of overstock and stock shortages indicates inventory allocation inefficiencies.

Recommendation

Management should implement inventory optimization initiatives, including:

- Improved demand forecasting.
- Inventory rebalancing between stores.
- Reduction of excess stock levels.
- Enhanced replenishment planning.
- Regular monitoring of inventory risk indicators.

Inventory management should be treated as the highest operational priority due to its significant contribution to overall business risk.

10.4 Finding 3: Revenue Targets Are Not Being Achieved

Observation

Revenue Risk was calculated at 57.97%. Quota Attainment was only 42.03%, indicating that actual sales performance remains substantially below planned revenue expectations.

Revenue gap analysis identified several underperforming stores and products contributing disproportionately to overall revenue risk.

Business Implications

Persistent revenue underperformance may reduce profitability, limit growth opportunities, and negatively impact future business investments.

Store-level and product-level underperformance suggest that revenue challenges are concentrated within specific business segments rather than uniformly distributed across the organisation.

Recommendation

Management should:

- Investigate root causes of underperforming stores.
- Review product portfolio performance.
- Strengthen sales planning processes.

- Evaluate promotional effectiveness.
- Implement targeted improvement initiatives for high-risk stores and products.
- Revenue performance should remain a strategic priority due to its direct impact on financial results.

10.5 Finding 4: Fulfilment Performance Requires Improvement

Observation

Fulfilment Risk was calculated at 47.33%. The analysis identified substantial levels of SLA breaches and delayed deliveries. Carrier delays emerged as the primary operational cause of fulfilment failures.

The majority of fulfilment issues were associated with delayed delivery tiers and standard shipping methods.

Business Implications

Poor fulfilment performance may negatively affect customer satisfaction, increase complaint volumes, and reduce customer loyalty.

Persistent delivery failures can also damage brand reputation and reduce repeat purchasing behaviour.

Recommendation

Management should:

- Review logistics and carrier performance.
- Establish stronger delivery performance monitoring.
- Investigate recurring delay causes.
- Improve operational coordination across fulfilment processes.
- Introduce service-level performance reviews for logistics providers.

Reducing fulfilment failures will improve both operational efficiency and customer experience.

10.6 Finding 5: Customer Risk Remains Moderate

Observation

Customer Risk was calculated at 28.63%. Customer Satisfaction (CSAT) averaged approximately 4.02, while Net Promoter Score (NPS) reached 62.77%, indicating generally positive customer sentiment.

Although customer risk remains lower than other risk domains, survey responses still indicate opportunities for improvement. It should be noted that customer survey data was available for only six countries and therefore does not represent the entire global customer base.

Business Implications

While overall customer sentiment appears positive, localised dissatisfaction may still exist within specific customer groups or regions.

Failure to address emerging customer concerns may increase future customer attrition risk.

Recommendation

Management should:

- Continue monitoring customer satisfaction trends.
- Investigate lower-performing regions.
- Expand customer feedback collection programs.
- Address recurring service issues affecting customer experience.
- Align customer improvement initiatives with fulfilment performance improvements.
- Maintaining customer satisfaction is essential for long-term business sustainability.

10.7 Summary

The Retail Risk Intelligence Center successfully identified critical business risks across multiple operational domains. Inventory management was identified as the most significant source of risk, followed by revenue performance and fulfilment operations. While customer sentiment remained relatively positive, opportunities for improvement still exist.

The dashboard provides management with a centralised risk-monitoring platform capable of supporting proactive decision-making, operational improvements, and continuous performance monitoring.

11. Assumptions and Limitations

11.1 Project Assumptions

Several assumptions were made during the development of the Retail Risk Intelligence Center to ensure consistency of analysis and support the construction of risk-based KPIs.

Single Reporting Currency

Validation confirmed that only one CurrencyKey existed across the primary transactional datasets. Therefore, all financial measures, including Sales, Sales Quota, Inventory Value, Gross Profit, and Refund Amount, were treated as a single reporting currency, and no foreign exchange conversions were applied.

Inventory Snapshot Assumption

The Inventory dataset was identified as a periodic snapshot dataset rather than a transactional inventory movement dataset. Inventory calculations were therefore designed using snapshot-based logic and point-in-time inventory values.

Equal Risk Weighting

The Overall Risk Score was calculated using equal weighting across the four risk domains:

- Revenue Risk (25%)
- Inventory Risk (25%)
- Customer Risk (25%)
- Fulfilment Risk (25%)

This approach was adopted to ensure balanced representation of all business functions within the overall risk framework.

Fixed Risk Thresholds

Risk classifications were based on predefined threshold ranges:

Risk Score	Level of Risk
0–25%	Low
25–50%	Moderate
50–75%	High
>75%	Critical

These thresholds were established to simplify interpretation and facilitate executive decision-making.

11.2 Project Limitations

Although the dashboard provides valuable business insights, several limitations should be considered when interpreting results.

Limited Customer Survey Coverage

Customer survey responses were available for only six countries within the dataset. Consequently, Customer Risk, Net Promoter Score (NPS), and Customer Satisfaction (CSAT) metrics represent surveyed markets only and should not be interpreted as a complete measure of global customer sentiment.

Historical Dataset

The Contoso dataset represents historical business data and does not reflect current real-world business conditions. Findings and recommendations are therefore intended for analytical demonstration purposes rather than operational implementation.

Absence of External Factors

The analysis was based solely on the data available within the Contoso dataset. External factors such as economic conditions, competitor activity, inflation, supply chain disruptions, and seasonal market events were not incorporated into the risk calculations.

Simplified Risk Framework

The Overall Risk Score was developed using a custom risk methodology designed for this project. While the framework provides a useful consolidated view of business risk, organizations may choose alternative weighting structures and risk models based on their specific strategic priorities.

Data Availability Constraints

Certain business processes contained limited historical records or incomplete geographic coverage. As a result, some analyses may not fully represent all operational scenarios.

11.3 Future Enhancements

Several opportunities exist to further enhance the Retail Risk Intelligence Center.

Potential future developments include:

- Implementation of predictive risk forecasting models.
- Integration of machine learning techniques for anomaly detection.
- Expansion of customer survey coverage across additional countries.
- Inclusion of external economic and market indicators.
- Development of automated alerting mechanisms for emerging risks.
- Real-time data integration and dashboard refresh capabilities.

These enhancements would further strengthen the solution's ability to support proactive business decision-making and enterprise risk management.

12. Conclusion

The Retail Risk Intelligence Center was developed to provide a proactive and integrated approach to business risk monitoring using the Microsoft Contoso Retail Dataset. The project successfully transformed raw operational, financial, customer, and fulfilment data into a structured analytical framework capable of identifying and monitoring organisational risk. Through the application of Python, SQL Server, Power BI, and DAX, multiple business processes were consolidated into a unified reporting solution.

The final dashboard enables users to monitor risk exposure across four key business domains:

- Revenue Risk
- Inventory Risk
- Customer Risk
- Fulfillment Risk

A custom KPI framework and Overall Risk Score methodology were developed to provide executives with a consolidated view of business performance and operational vulnerabilities. Analysis revealed that Inventory Risk represented the most significant source of organisational exposure, followed by Revenue Risk and Fulfilment Risk. Customer Risk remained comparatively lower but highlighted opportunities for continuous improvement.

The solution demonstrates the practical application of data analytics techniques, including data preparation, validation, dimensional modelling, KPI development, business intelligence reporting, and data storytelling. Furthermore, the project illustrates how data can be transformed into actionable insights that support strategic decision-making and operational improvement.

Overall, the Retail Risk Intelligence Center provides a scalable foundation for enterprise risk monitoring and showcases the end-to-end analytical workflow required in modern Data Analyst roles.

Appendix

Appendix A. Python Code Used to Convert Parquet Files to CSV Files

```
: import pandas as pd
import glob
import os

# Folder containing parquet files
folder = r"C:\Users\user\Downloads\contoso"

# Get all parquet files
parquet_files = glob.glob(os.path.join(folder, "*.parquet"))

# Convert each parquet file to csv
for file in parquet_files:
    df = pd.read_parquet(file)

    filename = os.path.basename(file)

    # Extract name after "_v" and change extension to .csv
    csv_name = filename.split("_v")[1].replace(".parquet", ".csv")

    csv_file = os.path.join(folder, csv_name)

    df.to_csv(csv_file, index=False)

    print(f"Converted: {filename} -> {csv_name}")

print("All files converted successfully!")
```

Converted: dim_vAcquisitionChannel.parquet -> AcquisitionChannel.csv
Converted: dim_vChannel.parquet -> Channel.csv
Converted: dim_vCurrency.parquet -> Currency.csv
Converted: dim_vCustomer.parquet -> Customer.csv
Converted: dim_vDate.parquet -> Date.csv
Converted: dim_vEmployee.parquet -> Employee.csv
Converted: dim_vPaymentMethod.parquet -> PaymentMethod.csv
Converted: dim_vProduct.parquet -> Product.csv
Converted: dim_vPromotion.parquet -> Promotion.csv
Converted: dim_vReturnReason.parquet -> ReturnReason.csv
Converted: dim_vStore.parquet -> Store.csv
Converted: fact_vCustomerAcquisition.parquet -> CustomerAcquisition.csv
Converted: fact_vCustomerSurvey.parquet -> CustomerSurvey.csv
Converted: fact_vExchangeRate.parquet -> ExchangeRate.csv
Converted: fact_vInventory.parquet -> Inventory.csv
Converted: fact_vMarketingSpend.parquet -> MarketingSpend.csv
Converted: fact_vOnlineSales.parquet -> OnlineSales.csv
Converted: fact_vOrderFulfillment.parquet -> OrderFulfillment.csv
Converted: fact_vOrderPayment.parquet -> OrderPayment.csv
Converted: fact_vReturns.parquet -> Returns.csv
Converted: fact_vSalesQuota.parquet -> SalesQuota.csv
Converted: fact_vStoreSales.parquet -> StoreSales.csv
All files converted successfully!

Appendix B. SQL Queries and Results of Datasets Completeness Validation

Queries	Result																																																
<pre>SELECT 'StoreSales' AS TableName, COUNT(*) AS RecordCount FROM StoreSales UNION ALL SELECT 'SalesQuota', COUNT(*) FROM SalesQuota UNION ALL SELECT 'Returns', COUNT(*) FROM Returns UNION ALL SELECT 'OrderPayment', COUNT(*) FROM OrderPayment UNION ALL SELECT 'OrderFulfillment', COUNT(*) FROM OrderFulfillment UNION ALL SELECT 'OnlineSales', COUNT(*) FROM OnlineSales UNION ALL SELECT 'MarketingSpend', COUNT(*) FROM MarketingSpend UNION ALL SELECT 'Inventory', COUNT(*) FROM Inventory UNION ALL SELECT 'CustomerSurvey', COUNT(*) FROM CustomerSurvey UNION ALL SELECT 'ExchangeRate', COUNT(*) FROM ExchangeRate UNION ALL SELECT 'AcquisitionChannel', COUNT(*) FROM AcquisitionChannel UNION ALL SELECT 'Channel', COUNT(*) FROM Channel UNION ALL SELECT 'Currency', COUNT(*) FROM Currency UNION ALL SELECT 'Customer', COUNT(*) FROM Customer UNION ALL SELECT 'CustomerAcquisition', COUNT(*) FROM CustomerAcquisition UNION ALL SELECT 'Date', COUNT(*) FROM Date UNION ALL SELECT 'Employee', COUNT(*) FROM Employee UNION ALL SELECT 'PaymentMethod', COUNT(*) FROM PaymentMethod UNION ALL SELECT 'Product', COUNT(*) FROM Product UNION ALL SELECT 'Promotion', COUNT(*) FROM Promotion UNION ALL SELECT 'ReturnReason', COUNT(*) FROM ReturnReason UNION ALL SELECT 'Store', COUNT(*) FROM Store ORDER BY TableName;</pre>	<table><tr><th>Results</th><th>Messages</th></tr><tr><th>TableName</th><th>RecordCount</th></tr><tr><td>1 AcquisitionChannel</td><td>7</td></tr><tr><td>2 Channel</td><td>4</td></tr><tr><td>3 Currency</td><td>28</td></tr><tr><td>4 Customer</td><td>18869</td></tr><tr><td>5 CustomerAcquisition</td><td>18484</td></tr><tr><td>6 CustomerSurvey</td><td>16773</td></tr><tr><td>7 Date</td><td>2556</td></tr><tr><td>8 Employee</td><td>287</td></tr><tr><td>9 ExchangeRate</td><td>773</td></tr><tr><td>10 Inventory</td><td>8013099</td></tr><tr><td>11 MarketingSpend</td><td>220</td></tr><tr><td>12 OnlineSales</td><td>12627608</td></tr><tr><td>13 OrderFulfillment</td><td>1674320</td></tr><tr><td>14 OrderPayment</td><td>1674320</td></tr><tr><td>15 PaymentMethod</td><td>7</td></tr><tr><td>16 Product</td><td>2517</td></tr><tr><td>17 Promotion</td><td>28</td></tr><tr><td>18 ReturnReason</td><td>8</td></tr><tr><td>19 Returns</td><td>581109</td></tr><tr><td>20 SalesQuota</td><td>7465911</td></tr><tr><td>21 Store</td><td>306</td></tr><tr><td>22 StoreSales</td><td>3406089</td></tr></table>	Results	Messages	TableName	RecordCount	1 AcquisitionChannel	7	2 Channel	4	3 Currency	28	4 Customer	18869	5 CustomerAcquisition	18484	6 CustomerSurvey	16773	7 Date	2556	8 Employee	287	9 ExchangeRate	773	10 Inventory	8013099	11 MarketingSpend	220	12 OnlineSales	12627608	13 OrderFulfillment	1674320	14 OrderPayment	1674320	15 PaymentMethod	7	16 Product	2517	17 Promotion	28	18 ReturnReason	8	19 Returns	581109	20 SalesQuota	7465911	21 Store	306	22 StoreSales	3406089
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Appendix C. SQL Queries and Results of Single Currency Validation

1	SELECT DISTINCT CurrencyKey
2	FROM StoreSales
3	ORDER BY CurrencyKey;
4	
5	SELECT DISTINCT CurrencyKey
6	FROM SalesQuota
7	
8	UNION
9	
10	SELECT DISTINCT CurrencyKey
11	FROM StoreSales
12	
13	UNION
14	
15	SELECT DISTINCT CurrencyKey
16	FROM OnlineSales;

121 % No issues found

Results	Messages
CurrencyKey	
1 1	

CurrencyKey	
1 1	

Appendix D. SQL Queries and Results of Product Validation

```
1 SELECT p.ProductKey,
2       p.ProductName
3 FROM Product p
4 LEFT JOIN StoreSales s
5       ON p.ProductKey = s.ProductKey
6 WHERE s.ProductKey IS NULL;
```

1 % No issues found

Results Messages

	ProductKey	ProductName
1	944	A. Datum Compact Digital Camera M200 Black

Appendix E. SQL Queries and Results of Inventory Structure Validation

```
SELECT COLUMN_NAME, DATA_TYPE
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'Inventory';

SELECT
    MIN(DateKey) AS FirstDate,
    MAX(DateKey) AS LastDate,
    COUNT(*) AS TotalRecords
FROM Inventory;

SELECT
    StockStatus,
    COUNT(*) AS RecordCount
FROM Inventory
GROUP BY StockStatus
ORDER BY RecordCount DESC;
```

Results Messages

	COLUMN_NAME	DATA_TYPE
1	InventoryKey	int
2	DateKey	date
3	InventoryDate	date
4	StoreKey	int
5	ProductKey	int
6	CurrencyKey	int
7	OnHandQuantity	int
8	OnOrderQuantity	int
9	SafetyStockQuantity	int
10	InventoryValue	float
11	OnOrderValue	float
12	StockCoverageRa...	float
13	StockStatus	nvarchar
14	DaysInStock	int
15	MinDayInStock	int
16	MaxDayInStock	int
17	Aging	int
18	AgingTier	nvarchar
19	IsCurrentSnapshot	bit

	FirstDate	LastDate	TotalRecords
1	2023-01-06	2025-12-...	8013099

	StockStatus	RecordCount
1	In Stock	3058932
2	Overstocked	2650523
3	Severely Overstock...	1189703
4	Low Stock	1100711
5	Stock-Out	13230

Appendix F. SQL Queries and Results of Customer Survey Country Validation

1	SELECT
2	c.CustomerCountry,
3	COUNT(*) AS SurveyResponses
4	FROM CustomerSurvey cs
5	INNER JOIN Customer c
6	ON cs.CustomerKey = c.CustomerKey
7	GROUP BY c.CustomerCountry
8	ORDER BY SurveyResponses DESC;
9	

121 %
No issues found

ResultsMessages

	CustomerCountry	SurveyResponses
1	United States	6949
2	Australia	3190
3	United Kingdom	1775
4	Germany	1684
5	France	1679
6	Canada	1496